

Homework 3

Solutions must be **typed** and **submitted electronically**.

1. You have seen a 2-out-of-2 secret sharing scheme based on XOR. Generalize that scheme to a 3-out-of-3 scheme and prove that it is secure. The new scheme should just use XOR (and not Shamir secret sharing).
2. I used 2-out-of-10 Shamir secret sharing over $\mathbb{Z}_{11}$ to share a secret. Alice's share was (4,6) and Bob's was (7,3). Two shares should be enough to reconstruct the secret. So, what was the secret, and what were the other 8 shares? Show your work.